INCH-POUND
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SUPERSEDING
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DETAIL SPECIFICATION

RELEASE, CARGO PARACHUTE, AIRDROP, M-3

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type of cargo parachute release assembly consisting of a release mechanism, upper and lower suspension link assemblies, parachute connector assemblies and a timer assembly. The parachute connector assembly, timer assembly and rope spool assembly may be procured separately (see 6.4).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, Engineering Center, Attn: SSCNC-UTE, Natick, MA 01760-5017 by using the Standardization Document Improvement Proposal (DD form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1670

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2. APPLICABLE DOCUMENTS

2.1. General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Other Government documents, drawings, and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S ARMY NATICK RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

11-1-3220 - Release Assembly, Cargo, Parachute, Airdrop, M-3
11-1-3230 - Spool Assembly, Rope
11-1-3236 - Toggle
11-1-3240 - Connector Assembly, Parachute
11-1-3246 - Housing, Front and Rear (Casting)
11-1-3250 - Timer Assembly, Parachute

(Copies of drawings are available from the U.S.Army Natick Research, Development and Engineering Center, ATTN: SSCNC-IMD, Natick,MA 01760-5017).

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.3).

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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B46.1 - Surface Texture (Surface roughness, Waviness, and Lay)

(Applications for copies should be addressed to the American National Standards Institute, Inc. 11 West 42nd Street, New York, NY 10036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 1444 *Standard Practice for Magnetic Particle Examination*

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.1) in accordance with 4.2.

3.2 Material. The contractor shall select the material, but the material shall be capable of meeting all of the operational and environmental requirements specified herein. The material specified on Drawing 11-1-3220, and all subsidiary drawings and parts lists pertaining thereto, is recommended, but is not mandatory.

3.3 Construction and supplementary process. The construction and supplementary process used in the manufacture of the cargo parachute releases shall conform to the requirements specified or shown on drawing 11-1-3220, and all subsidiary drawings and parts lists pertaining thereto, and as specified herein.

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3.4 Surface finish. American National Standards Institute (ANSI) Standard No. B46.1, shall apply for the interpretation of surface roughness values specified on the applicable drawings.

3.5 Performance.

3.5.1 Cargo parachute connector proofload. The cargo parachute connector shall show no visual evidence of failure (such as a fracture or permanent deformation) when subjected to a tensile proofload of 45,000 pounds for 15 seconds when tested as described in 4.12.3.

3.5.2 Release assembly proofload. The release assembly shall show no visual evidence of failure (such as a fracture or permanent deformation) when subjected to a tensile proofload of 135,000 pounds for 15 seconds when tested as described in 4.11.4.

3.5.3 Release assembly unlocking. At completion of the timer unwinding cycle, the timer shall retract the toggle lock slides to the unlocked position with no evidence of interference when tested as specified in 4.11.4.

3.5.4 Release assembly, functional. The release assembly shall show no evidence of interference when subjected to the no load testing as specified in 4.11.5.

3.5.5 Delay timing cycle. The delay timing cycle shall be 25 seconds,plus or minus 3 seconds, from start to completion. There shall be no evidence of interference in retracting the toggle lock slides to the unlocked position after completion of the unwinding cycle when tested as specified in 4.11.1.

3.5.6 Delay timer retraction arms. When functionally tested as specified in 4.11.2, the retraction arms shall rotate through an angle of 23 degrees during the timing cycle. At completion of the timing cycle, the arms shall retract to the unwound position, tangent to the timer housing surface. Suggested reference drawing 11-1-3250 indicates the position and tolerances of the retraction arms movement.

3.5.7 Government furnished property. The following items, when required for proofload testing of the release assembly, separately procured cargo parachute connectors or timer assemblies, may be obtained from the procuring activity on a loan basis (see 6.3).

Release, Cargo Parachute, Airdrop, M-3; Part No. 11-1-3220-1

Timer Assembly; Part No. 11-1-3250-1

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3.5.8 Identification marking. Identification marking shall be in accordance with the legends and locations as shown on the referenced drawings.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

4.2 First article inspection. First article inspection shall be performed on one complete release assembly if a first article sample is required (see 3.1). This inspection shall include the examinations of 4.3.

4.3 Examination. Each release assembly shall be examined for compliance with the requirements specified in 3.2 through 3.5. This element of inspection shall encompass all visual examinations (4.9) and dimensional measurements (4.10) and the tests of (4.11). Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection of the lot.

4.4 End item visual examination. The end items shall be examined for the defects listed in Table I. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection of the lot.

4.5 Components and material inspection. In accordance with 4.x, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.6 In-process inspection. Inspection of subassemblies shall be made to ascertain that construction details which cannot be examined in the finished product are in accordance with specified requirements. Whenever nonconformance is noted, correction shall be made to affected items and process. Items that cannot be corrected shall be removed from the process. The Government reserves the right to exclude from consideration for acceptance, any material or service for which in-process inspection has indicated nonconformance.

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4.6.1 In-process dimensional examination. Prior to assembly, but after heat treatment and application of protective finish, the toggle and the front and rear housing shall be examined for conformance to the dimensions indicated on their respective drawings. The finding of any defect shall be cause for the rejection of the lot.

4.7 Heat treatment inspection and hardness tests. Heat treated parts shall be inspected to determine compliance with requirements specified on the applicable drawings. The finding of any defect shall be cause for rejection of the lot.

4.8 Magnetic particle inspection. Each release assembly shall be inspected as specified in accordance with ASTM E 1444 for determination of compliance with applicable drawing requirements.

4.9 End item visual examination. The end items, defined as completely fabricated units of the Release Assembly being purchased, shall be examined for the defects listed in table I. The lot size shall be expressed in units of Release Assemblies. The sample unit shall be one release assembly. The finding of any defect shall be cause for the rejection of the lot.

TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor
Finish	Omitted or not type specified	101	
	Coating powdery, coarse grained, mottled, not adherent, evidence of corrosion	102	
	Supplementary treatment omitted, not type specified, tacky	103	
	Color not as specified		201
	Dirt, oil, grease or other foreign matter		202
Design	Any departure from design specified	104	
Construction and workmanship	Surface roughness value in excess of rating specified on drawing	105	
	Any component missing, damaged or misaligned	106	
	Sharp edges, burrs or metal slivers	107	

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Construction and workmanship	Components incorrectly assembled	108	
	Components loose	109	
	Movement of parts inhibited	110	
	Threaded section damaged	111	
Identification	Missing, incomplete, illegible improper location, or not as specified		203

4.10 End item dimensional inspection. The end item shall be examined for conformance to the dimensions specified on the drawings. Only those dimensions that can be evaluated without damaging or disassembling the end item shall be examined. Any dimensions not within the specified tolerance shall be classified as a defect. The sample unit shall be one release assembly. The finding of any defect shall be cause for rejection of the lot.

4.11 End item testing.

4.11.1 Release assembly and connector. Cargo parachute connectors and release assemblies (proofload, unlocking and functional) shall be tested as specified in 4.11.3, 4.11.4, and 4.11.5 for compliance with the requirements of 3.5.2, 3.5.3, 3.5.4, respectively. The lot size shall be no more than 50 completely assembled cargo parachute connectors or complete release assemblies. The sample unit shall be one completely assembled cargo parachute connector or complete release assembly. Cargo parachute connectors showing any visual evidence of failure (fracture or permanent deformation) shall be cause for rejection of the lot. Release assemblies showing any visual evidence of failure (fracture or permanent deformation) of any component of the release assembly, exclusive of the cargo parachute connectors and time delay assembly, or failure of the time delay assembly to retract the toggle lock slides freely and without interference to the unlocked position shall be cause for rejection of the lot.

4.11.2 Timer. The timer shall be tested as specified in 4.11.7 and 4.11.8 for compliance with the requirements of 3.5.5 and 3.5.6 respectively. The lot size shall be expressed in units of timers. The finding of any defect shall be cause for rejection of the lot.

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4.11.3 Cargo parachute connector proofload testing. The cargo parachute connector to be tested shall be installed in a contractor test fixture. The test fixture shall be attached to a direct reading device capable of imposing a 45,000 pound tensile force on the fixture and test item. A proofload of 45,000 pounds shall be applied to the connector at a rate of not less than 3,000 pounds/second nor greater than 6,000 pounds/second. The connector shall withstand the tensile load for 15 seconds after which the load shall be reduced to zero. The proofload of 45,000 pounds shall be immediately re-applied, held for 15 seconds, and again reduced to zero.

4.11.4 Release assembly proofload and unlocking testing. The release assembly shall be assembled with seven cargo parachute connectors and with a timer delay assembly (GFP,see 6.3), installed in the locked position. The locked position requires turning the timer stem approximately 90 degrees clockwise to engage the toggle lock slides with the toggle,inserting the arming wire through the hole in the timer stem. Rectangular holes are provided in the side plates of the release to ensure alignment of the toggles and toggle lock slides before turning the timer stem. The complete assembly shall then be mounted in a contractor furnished test fixture. The test fixture shall be attached to a calibrated tensile testing machine capable of imposing a 135,000 pound tensile force on the test fixture and release assembly. The fixture shall provide that the total applied tensile force be equally distributed among the seven cargo parachute connectors and also divided equally between the two lower link assemblies. The cargo parachute connector sleeves may be removed to facilitate attachment of the connector bodies to the test fixture. The lower link assemblies shall be connected so that only one branch of each link assembly will be subjected to the tensile load. The proofload of 135,000 pounds shall be applied to the release assembly at the rate of not less than 5,000 pounds/second, nor greater than 10,000 pounds/second. The release assembly shall withstand the tensile load for 15 seconds, after which the tensile force shall be reduced to zero. The proofload of 135,000 pounds shall be immediately re-applied, held for 15 seconds, then reduced to zero. After completion of the second proofload application, a tensile load of 500 pounds shall be applied to the release assembly. The release assembly shall then be activated by removing the arming wire. After the timing cycle is completed, the time delay assembly shall freely retract the toggle lock slides to the unlocked position. The tensile load shall the be reduced to zero and the release assembly removed from the test fixture and disassembled for inspection.

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4.11.5 Functional testing. After completion of the proof loading and timer unlocking testing, place the release assembly on a table with the timer and seven parachute connectors installed. Arm the release by turning the timer stem 90 degrees clockwise engaging the toggles and toggle lock slides and inserting the arming wire. Activate the timer by removing the arming wire. At the end of the time delay cycle and retraction of the toggle lock slides, the following sequence of events are to be performed without interference between the mating parts: rotate the upper suspension link to one side, fold down the upper retainer plate, and withdraw all seven parachute connectors from the release.

4.11.6 Temperature conditioning for timing cycle test. The temperature conditioning for the timer assembly shall be conducted in accordance with the following requirements:

Test Method	Requirements
High Temperature, Storage	Condition the timer at a steady state (non-cyclic) temperature of 160 degrees F (71 degrees C) for 48 hours with a relative humidity not exceeding 15%
Low Temperature Storage	Condition the timer at a temperature of -60 degrees F (-51 degrees C) for 24 hours

4.11.7 Timing cycle test. a) After high temperature conditioning, return the timer standard ambient conditions and assemble it into an M-3 release. Arm the timer by turning the timer stem approximately 90 degrees clockwise to engage the toggle lock slides with the toggles and insert the arming wire through the hole in the stem. Rectangular holes are provided in the side plates of the release to ensure alignment of the toggles and toggle lock slides before turning the timer stem. Activate the timer by removing the arming wire, and record the time delay.The timer shall freely and smoothly retract the toggle lock slides to the unlocked position. Failure of the assembly to meet either the time requirement of 3.5.5 or any indication of interference in retracting the toggle lock slides shall constitute a defect.

b) After low temperature conditioning, return the timer to standard ambient conditions, and assemble it into an M-3 release. Arm the timer by turning the timer stem approximately 90 degrees clockwise to engage the toggle lock slides with the toggles and insert the arming wire through the hole in the stem. Rectangular holes are provided in the side plates of the release to ensure alignment of the toggles and toggle lock slides before turning the timer stem. Activate the timer by removing the arming wire,

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4.11.7 Timing cycle test (cont'd) and record the time delay. The timer shall freely and smoothly retract the toggle lock slides to the unlocked position. Failure of the assembly to meet either the time requirement of 3.5.5 or any indication of interference in retracting the toggle lock slides shall constitute a defect.

4.11.8 Delay timer retraction arm test. The delay timer shall be wound so the retraction arms are in the extended position, and the angular travel of the arms shall be measured. At the end of the timing cycle the arms must move to their unwound position and the position of the arms shall be measured. Any deviation from the requirements specified in 3.5.6 shall constitute a defect.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.4). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Release, Cargo Parachute, Airdrop M-3, covered by this document is used to automatically disengage the parachutes from the load at the moment of ground impact. This prevents the parachutes from dragging or overturning the load. This release is intended for loads having a gross rigged weight of 60,000 pounds.

6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of Federal Acquisition Regulation (FAR) 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

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6.3 Government furnished property. The contracting Officer should arrange to furnish the property listed in 3.5.7.

6.4 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2 and 2.3).
c. When first article is required (see 3.1, 4.2, and 6.2).
d. Preservation and packing (see 5.1).

6.5 Subject term (key word) listing.

Airdrop System
Cargo loads
Parachute
Parachute Mechanism
Resupply

Custodians:
Army - GL
Navy - AS

Preparing Activity:
Army -GL
(Project 1670-0856)

Review Activities:
Army - AV

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-DTL-44481A	1996 MAY 15

3. DOCUMENT TITLE

RELEASE, CARGO PARACHUTE, AIRDROP, M-3

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON/DSN
U.S. Army Natick RD&E Center	(508) 233-4216	256-4216
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: SSCNC-UTE Natick, MA 01760-5017	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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